Experiment 4

Total Yearly Sales and Percentage Increase

Aim

To calculate the total sales for four quarters and determine the percentage increase in sales from the first quarter to the fourth quarter using NumPy arrays and arithmetic operations.

Procedure

1. Import the NumPy library.
2. Create a NumPy array named sales_data containing the sales amounts for four quarters.
3. Use np.sum(sales_data) to calculate the total yearly sales.
4. Extract the first-quarter sales using sales_data[0].
5. Extract the fourth-quarter sales using sales_data[3].
6. Calculate the percentage increase using the formula:

$$\text{Percentage Increase} = \frac{Q4 - Q1}{Q1} \times 100$$

7. Display the total yearly sales and the percentage increase.

Code

```
import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)

percentage = (
    (sales_data[3] - sales_data[0])
    / sales_data[0]
) * 100

print("Total yearly sales:", total_sales)

print("Percentage increase:", percentage, "%")
```

Result

Thus, the total sales for the year and the percentage increase from the first quarter to the fourth quarter were calculated successfully using NumPy arrays and arithmetic operations.

Output

```
========== RESTART: D:/fundamentails of data science/experiment 4.py ==========
Total yearly sales: 55000
Percentage increase: 80.0 %
```